Question 5

Circular motion · 5 marks DIFFICULTY: MEDIUM

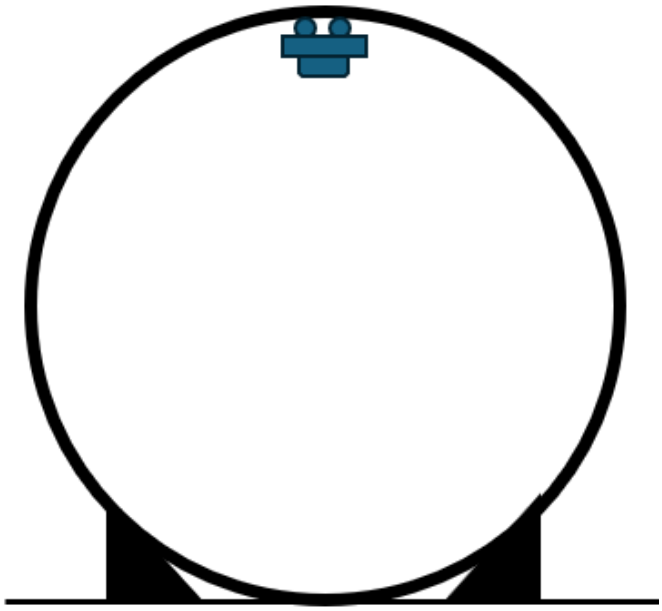


FIGURE 5

FIGURE 5 shows a toy car moving in anticlockwise direction at a constant speed of 3 $\mathrm{m\,s^{-1}}$ inside a vertical track with a 50 cm diameter. The mass of the toy car is 250 g. Assuming that the toy car is always in contact with the track, calculate the

(a) centripetal force.

(b) normal force when the toy car is at the top **AND** at the bottom of the track.

[5 marks]

Student working space

Answer scheme with step-by-step student hints

Radial-force convention: take the direction toward the centre of the circular track as positive.
Refer to the original free-body diagrams beside the force equations.

Given: $m = 250\ \text{g} = 0.250\ \text{kg}$, $d = 50\ \text{cm} = 0.50\ \text{m}$, $v = 3.0\ \text{m s}^{-1}$, $g = 9.81\ \text{m s}^{-2}$.

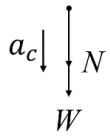
Step 1 (a) Determine the centripetal force GJU1

Student hint: Convert mass and diameter to SI units first. The radius is half the diameter; then use $F_c = mv^2/r$.

$$r = \frac{d}{2} = \frac{0.50}{2} = 0.25\ \text{m}$$
$$F_c = \frac{mv^2}{r} = \frac{(0.25)(3)^2}{0.25} = \boxed{9.00\ \text{N}}$$

Step 2 (b) At the top: resolve forces toward the centre G1

Student hint: The centre of the circle is downward at the top. From the extracted FBD, both N and $W = mg$ act toward the centre.



Original top FBD

$$F_c = N + W$$
$$N = F_c - mg$$

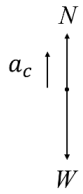
Step 3 Calculate the normal force at the top JU1

Student hint: Subtract the weight from the centripetal-force magnitude.

$$N_{\text{top}} = 9 - (0.25)(9.81) = \boxed{6.55\ \text{N}}$$

Step 4 (b) At the bottom: resolve forces toward the centre G1

Student hint: The centre is upward at the bottom. The original FBD shows N upward and $W = mg$ downward. Thus their net upward force provides F_c.



Original bottom FBD

$$F_c = N - W$$
$$N = F_c + mg$$

Step 5 Calculate the normal force at the bottom JU1

Student hint: Add the weight to the centripetal-force magnitude.

$$N_{\text{bottom}} = 9 + (0.25)(9.81) = \boxed{11.45\ \text{N}}$$

Total: 5 marks